

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

M1a: Data and Sampling

Prof. Oke

UMassAmherst

College of Engineering

September 8, 2026

Outline

- ① Probability, Statistics and Uncertainty
- ② Frequency and Histograms
- ③ Outlook

Welcome

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Please take the time to do the following:

- Fill out the class pre-survey at <https://forms.gle/8Am5m39bxoUhRzt19>

Probability and statistics

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In summary, probability theory enables us to find the consequences of a given ideal world, whereas, statistical theory permits us to measure the extent to which our world is ideal.

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Probability and statistics in engineering

In engineering, probability and statistics enable us to:

- Describe phenomena
- Develop robust design and decision making procedures
- Quantify uncertainty and risk

Uncertainty in engineering

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Helpful to consider 2 broad categories of uncertainty:

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- Due to imperfect representations (idealized models)
- Resulting in inaccurate estimates e.g. central values (mean, median), etc

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Value	Probability	
	Model A	Model B
1	$1/6$	0.1
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4	$1/6$	0.2
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Both models represent *epistemic uncertainty* in the properties of the die. With additional data (further rolls), the correctness of the two models can be accurately judged.

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The *aleatory uncertainty* remains the same, but our quantification can improve with more information.

Activity 1: Birthday Paradox

- Before we do anything else, I want everyone to make a prediction. We have 160 people in this class. What do you think is the probability that at least two people in this room share the exact same birthday (same month and day)?

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- Before we do anything else, I want everyone to make a prediction. We have 160 people in this class. What do you think is the probability that at least two people in this room share the exact same birthday (same month and day)?
- Write down a percentage from 0% to 100%. Be honest about your first instinct—there are no wrong answers here!

Example 2: Rainfall intensity

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Year No.	Rainfall Intensity, in.	Year No.	Rainfall Intensity, in.	Year No.	Rainfall Intensity, in.
1	43.30	11	54.49	21	58.71
2	53.02	12	47.38	22	42.96
3	63.52	13	40.78	23	55.77
4	45.93	14	45.05	24	41.31
5	48.26	15	50.37	25	58.83
6	50.51	16	54.91	26	48.21
7	49.57	17	51.28	27	44.67
8	43.93	18	39.91	28	67.72
9	46.77	19	53.29	29	43.11
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6	50.51	16	54.91	26	48.21
7	49.57	17	51.28	27	44.67
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Plot a histogram of the data

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Interval	No. of Observations	Fraction of Total Observations
38–42	3	0.1034
42–46	7	0.2415
46–50	5	0.1724
50–54	5	0.1724
54–58	3	0.1034
58–62	3	0.1034
62–66	1	0.0345
66–70	2	0.0690

Activity 2: Data About Us

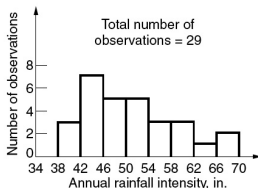
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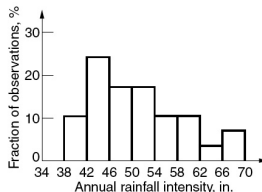
- Plot counts (or fraction) on the *ordinate* (y) axis and intervals on the *abscissa* (x)

Example 2: Rainfall intensity (cont.)

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(a) In number of observations



(b) In fraction of total observations

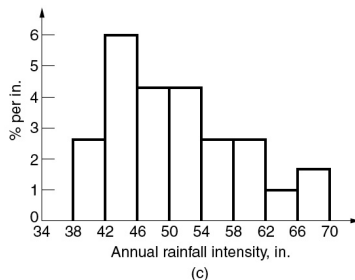
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For further analyses, comparing the empirical frequency to a theoretical frequency distribution is necessary. In such cases, we find the **empirical frequency function** by obtaining a frequency diagram with an **area of 1**.

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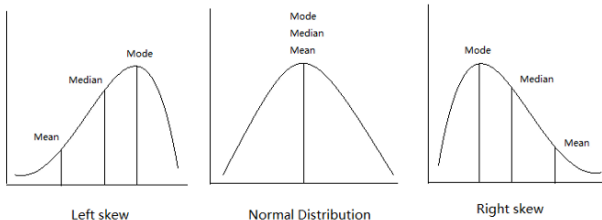
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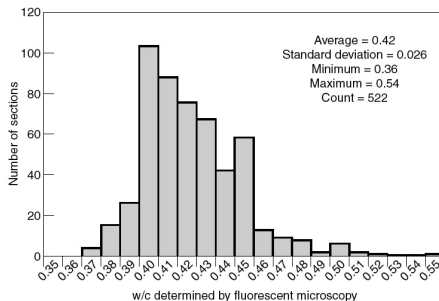
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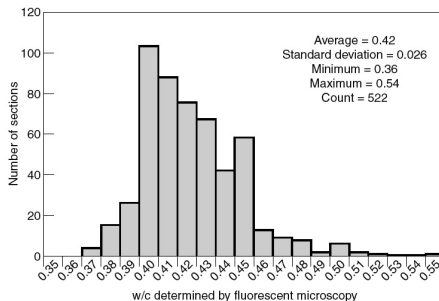
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Example 3: Water-cement ratio of concrete specimens

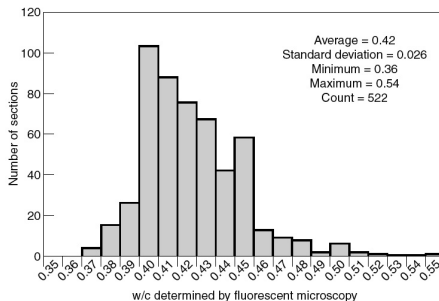
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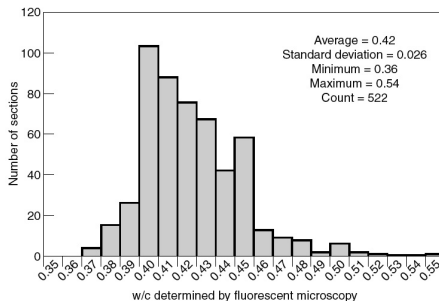


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A lower ratio indicates greater stiffness and strength

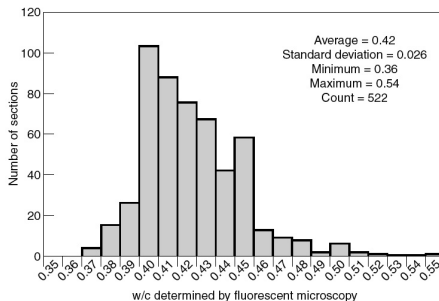
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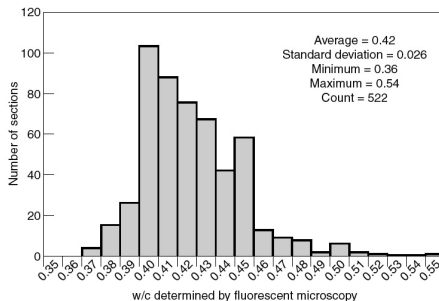


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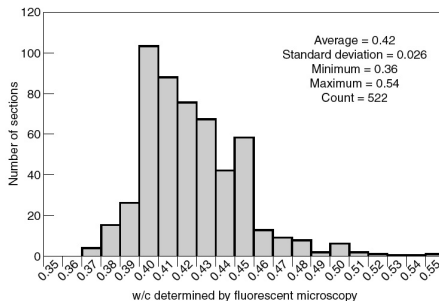


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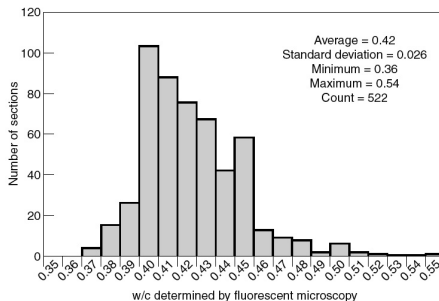


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- What is the mode of this sample?
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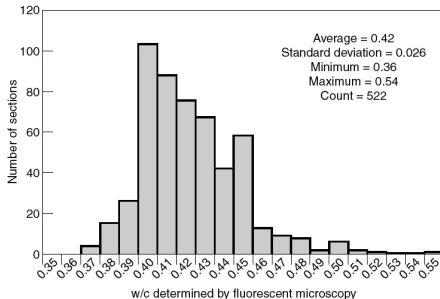


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- What is the mode of this sample? 0.39
- What is the *range* of this sample? $\text{Range} = 0.54 - 0.36 = 0.18$

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Is this distribution left-skewed or right-skewed?

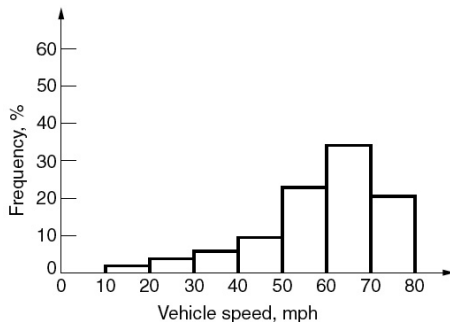


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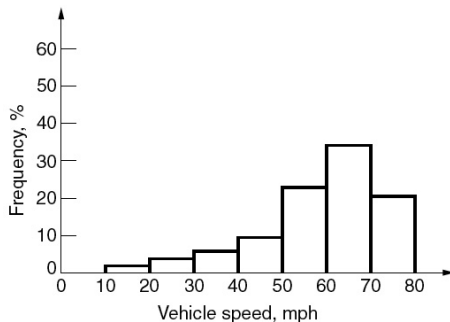


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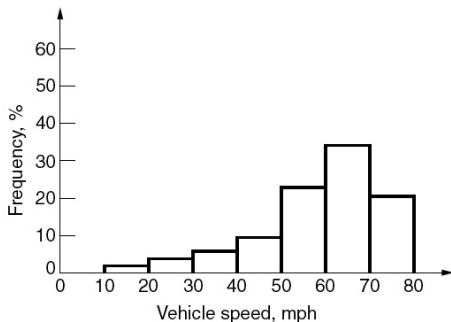


Figure 1 18 Impact speeds of passenger car accidents

This distribution is left-skewed (long left tail; mean left of mode)

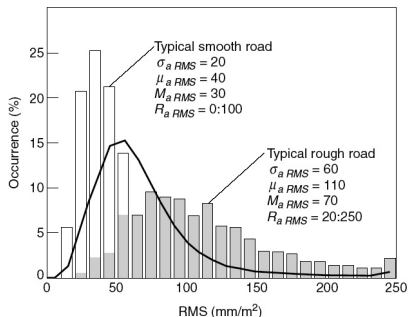
Example 5: Road roughness profiles

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Surface roughness is measured by the root mean square (RMS) of height deviations per given area. Here, we are given the distributions for two samples and fitted *mixture distribution*. σ is the standard deviation, μ is the mean.

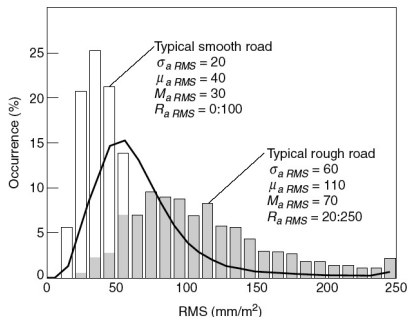
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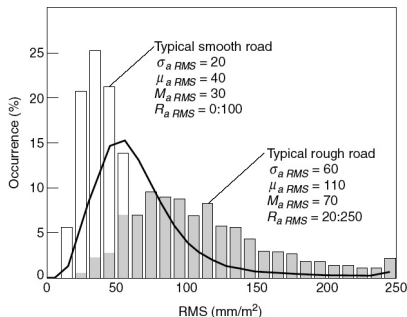
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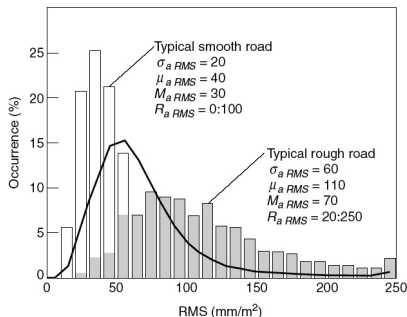


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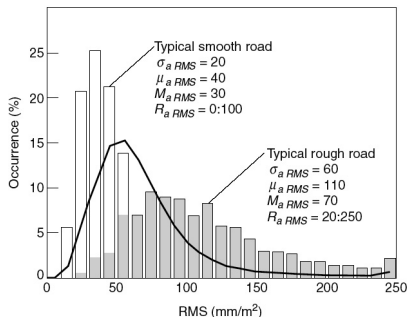


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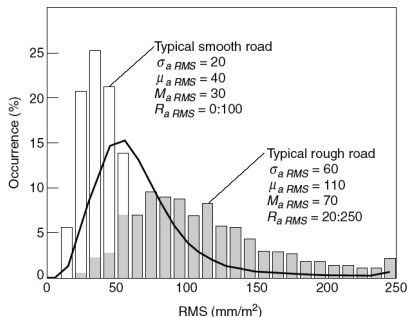


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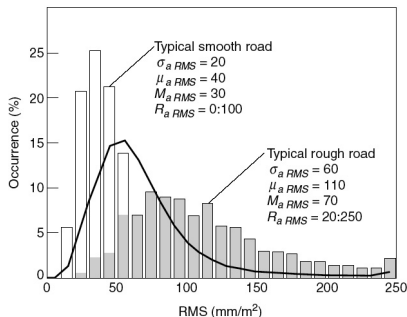


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- What does M represent? **Mode**
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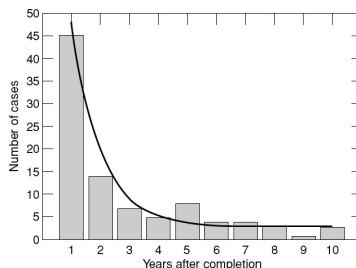
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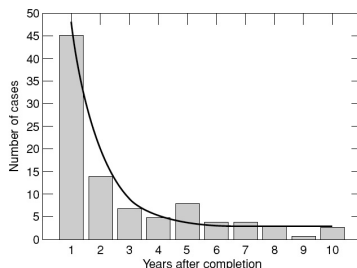
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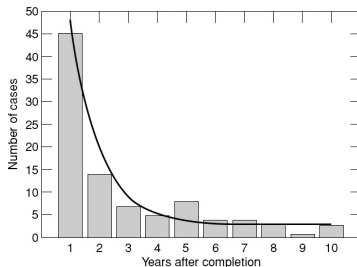
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What is the associated theoretical PDF of the histogram in the above figure?

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- Uncertainty in engineering

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- Histograms and scattergrams

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